OPTIMIZED PERFORMANCE

RESEARCH PEPTIDE SUPPLY

Business Status Report

April 6, 2026

Target Launch: May 19, 2026

CONFIDENTIAL — INTERNAL USE ONLY

Executive Summary: Optimized Performance is a direct-to-consumer research peptide company. The e-commerce platform is built on Next.js 14 / React 18, deployed on Vercel, with MoonPay cryptocurrency payment processing (fiat → USDC on Polygon). The website, admin dashboard, product catalog, and payment rail are functional. Key remaining work includes environment configuration, supplier sourcing, MoonPay KYB verification, and legal review before the May 19 launch.

1. PLATFORM ARCHITECTURE

TECH STACK

- **Framework:** Next.js 14.2 + React 18.3
- **Hosting:** Vercel (auto-deploy from GitHub)
- **Payment:** MoonPay (fiat → USDC/Polygon)
- **Database:** Supabase (PostgreSQL) — configured, not yet provisioned
- **Cache/Store:** Upstash Redis — referenced, not yet provisioned
- **Alerts:** SendGrid (email) + Twilio (SMS) — modules built, not configured
- **Styling:** CSS-in-JS inline (no framework)
- **Repo:** GitHub — matthewredfearn75-cmd/optimized-performance-site

PAGES & ROUTES

- **/** — Landing page with hero, featured products, trust badges
- **/shop** — Full product catalog grid
- **/checkout** — Shipping form + MoonPay payment overlay
- **/admin** — Password-protected admin dashboard
- **/privacy** — Privacy policy
- **/terms** — Terms of service + RUO disclaimer
- **/api/*** — Inventory, orders, webhooks endpoints

2. PRODUCT CATALOG

PRODUCT	DOSAGE	FORMAT	PRICE	VIAL LABEL	IN CATALOG
BPC-157 — Body Protection Compound	5 mg	Lyophilized / 2 mL vial	$39.99	Ready	Yes
TB-500 — Thymosin Beta-4 Fragment	5 mg	Lyophilized / 2 mL vial	$34.99	Ready	Yes
PT-141 — Bremelanotide	10 mg	Lyophilized / 2 mL vial	$44.99	Ready	Yes
CJC-1295 — Mod GRF (No DAC)	5 mg	Lyophilized / 2 mL vial	$42.99	Ready	Yes
Ipamorelin — GH Secretagogue	5 mg	Lyophilized / 2 mL vial	$36.99	Ready	Yes
GHRP-2 — GH Releasing Peptide-2	5 mg	Lyophilized / 2 mL vial	$29.99	Ready	Yes
GLP-3 — Triple Agonist Peptide (retatrutide)	10 mg	Lyophilized / 2 mL vial	TBD	Ready	Not yet
GLP-3 — Triple Agonist Peptide (retatrutide)	20 mg	Lyophilized / 2 mL vial	TBD	Ready	Not yet

Note: GLP-3 is branded to avoid Eli Lilly trademark issues. Both 10 mg and 20 mg labels are created and ready for thermal printing (0.5" x 1.5" metallic labels). GLP-3 still needs to be added to `products.js` with pricing and added to the shop page.

3. ADMIN DASHBOARD

Accessible at **/admin** — password-protected (default: `optimized2024`, configurable via `ADMIN_PASSWORD` env var).

TAB	FEATURES	STATUS
Orders	Status pipeline (Pending → Packed → Shipped → Fulfilled), create orders, expandable detail view, tracking # entry, CSV export, auto-decrements supply on fulfillment	Built
Inventory	Editable stock quantities per SKU, total value calculation, low/out-of-stock alerts, saves to Supabase	Built
Supply Tracker	Lot number management (auto-generate or manual), supplier lot tracking, COA status, vial counts, date received	Built

Architecture note: The admin page was refactored from a single 737-line file into 4 focused component files for maintainability. Each tab is its own component (~180–430 lines).

4. PAYMENT INFRASTRUCTURE

> **Payment Flow:** Customer pays with credit/debit card → MoonPay converts fiat to USDC (stablecoin on Polygon blockchain) → ~4% processing fee → funds arrive in configured wallet. MoonPay handles KYC for customers automatically.

- **SDK:** @moonpay/moonpay-react v1.10.6 — integrated as overlay widget in checkout
- **Webhook:** `/api/webhooks/moonpay` — verifies HMAC signature, updates order to "completed", decrements inventory, sends confirmation email
- **Current state:** Using test/placeholder API key — needs production key + KYB verification with MoonPay
- **Off-ramp:** USDC → fiat conversion and bank withdrawal setup still needed

5. VIAL LABELS

All labels are designed, generated, and stored in the `labels/` directory. Each label is available as both SVG (source) and JPEG (300 DPI, 1875x625px) for thermal printer use.

FILE	PRODUCT	DOSAGE	DIMENSIONS
bpc-157-vial-label.jpg	BPC-157	5 mg	0.5" x 1.5"
tb-500-vial-label.jpg	TB-500	5 mg	0.5" x 1.5"
pt-141-vial-label.jpg	PT-141	10 mg	0.5" x 1.5"
cjc-1295-vial-label.jpg	CJC-1295	5 mg	0.5" x 1.5"
ipamorelin-vial-label.jpg	Ipamorelin	5 mg	0.5" x 1.5"
ghrp-2-vial-label.jpg	GHRP-2	5 mg	0.5" x 1.5"
glp-3-vial-label.jpg	GLP-3	10 mg	0.5" x 1.5"
glp-3-20mg-vial-label.jpg	GLP-3	20 mg	0.5" x 1.5"

A generator script (`labels/generate-all-labels.js`) is included to create new labels for future products.

6. BRANDING

VISUAL IDENTITY

- **Logo:** Hexagonal mandala SVG (geometric, scientific feel)
- **Primary dark:** #0D1B2A (navy)
- **Primary accent:** #00B4D8 (cyan)
- **Secondary accent:** #0077B6 (darker cyan)
- **Typography:** Helvetica Neue / Arial (system fonts)
- **Design approach:** Minimalist, clean, card-based layouts

BRAND POSITIONING

- Research-grade peptides for scientific use
- Emphasis on purity (>98%), third-party testing
- "For Research Use Only" disclaimer throughout
- Trust signals: purity badges, secure checkout, fast shipping
- Crypto-native payment (appeals to privacy-conscious buyers)

7. WHAT'S DONE VS. WHAT'S REMAINING

AREA	ITEM	STATUS
Website	Home, Shop, Checkout pages	Done
Website	Privacy Policy + Terms of Service	Done
Website	Mobile responsiveness testing	Needs testing
Website	SEO meta tags + OG images	Not started
Admin	Orders, Inventory, Supply Tracker tabs	Done
Payment	MoonPay SDK integration	Done
Payment	MoonPay KYB verification (production)	Not started
Payment	Wallet + off-ramp to bank	Not started
Infra	Vercel deployment pipeline	Done
Infra	Supabase database provisioning	Not started
Infra	Environment variables configured	Not started
Infra	Custom domain + DNS	Not started
Infra	Email alerts (SendGrid)	Code ready, not configured
Products	6 peptides in catalog	Done
Products	GLP-3 added to catalog	Labels ready, not in products.js

Products	SKU + stock fields on products	Missing from data
Products	All 8 vial labels	Done
Operations	Supplier sourced	Unknown
Operations	Shipping carrier setup	Not started
Operations	Lot numbering system	Built into admin
Legal	Business entity (LLC)	Unknown
Legal	Legal review of Terms/Privacy	Not started

8. REPOSITORY STRUCTURE

PATH	PURPOSE	LINES
src/pages/index.js	Home / landing page	263
src/pages/shop.js	Product catalog	149
src/pages/checkout.js	Checkout + MoonPay	438
src/pages/admin/index.js	Admin shell + auth	231
src/pages/admin/OrdersTab.js	Order management	427
src/pages/admin/InventoryTab.js	Stock management	180
src/pages/admin/SupplyTab.js	Lot tracking	235
src/components/	Header, Footer, CartDrawer, ProductCard	686 total
src/lib/	supabase.js, alerts.js, inventory.js	188 total
src/data/products.js	Product catalog data	158
labels/	8 SVG + 8 JPEG vial labels + generator	—
Total source code		**~3,990**

9. GIT HISTORY SUMMARY

The repository has been developed across 2 branches:

- **master** — Production base: initial site, MoonPay integration, admin panel, API routes, legal pages, Supabase/Redis/alerts infrastructure

- **claude/review-peptide-website-pxuxm** — Current working branch: supply tracker, orders tab, vial labels, admin refactor, build fixes

PR #1 is open for merging the working branch into master. CI checks passing (Vercel Preview + GitGuardian).